

KUMAR VARENYA

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TECHNICAL SKILLS

Languages: C/C++, Python, Java, SQL, R, Bash, SystemVerilog, VHDL, ARM Assembly
Perception & Systems: OpenCV, Eigen, Kalman Filtering, Sensor Fusion, perf, Valgrind, GDB, CMake
Cloud & Infrastructure: AWS (CDK, Fargate, Bedrock, Comprehend), Docker, Kubernetes, Linux/Unix, Git
Data & Quant: PostgreSQL, PostGIS, Pandas, cvxpy, scikit-learn, Matplotlib, Jupyter, QGIS
Static Analysis & Testing: Clang Static Analyzer, ThreadSanitizer, GoogleTest, pytest

EXPERIENCE

Meta (Reality Labs)

Toronto, ON

Software Engineer

April 2026 – Present

- Implemented C++ visual-inertial odometry components for an inside-out headset tracking prototype, fusing camera and IMU streams with an extended Kalman filter.
- Built a Python/C++ evaluation harness that replays recorded sensor sessions against tracking builds and flags pose-drift regressions in CI.
- Profiled CPU and memory use on-device with perf and Valgrind, and cut per-frame processing time by roughly 10% on the tracking thread.

UBC Investment Management Trust Inc.

Vancouver, BC

Risk/Systems Engineer (UBC CS DMM: Operations Research & Financial Engineering)

September 2025 – April 2026

- Prototyped a mean-variance portfolio optimizer in Python (cvxpy) with turnover and sector-exposure constraints, tested on a subset of the endowment's public-equity sleeve.
- Built a scenario and stress-testing notebook that applied historical shocks (2008, 2020) to factor exposures, with results reviewed by the investment team.

Amazon

Vancouver, BC

Software Development Engineer Intern (AWS)

May 2024 – December 2024

- Built an internal LLM-powered summarization service on AWS (CDK, Fargate, Bedrock), containerized with Docker and deployed to a Kubernetes-backed environment.
- Designed prompt templates and added a PII-detection step (regex plus Amazon Comprehend) that cut sensitive-data leaks flagged in testing from 14 to 2.
- Wrote unit and integration tests for the service and shipped feature to production ahead of team's internal launch date.

Population Data BC

Vancouver, BC

Software Programmer

September 2022 – December 2025

- Built Python and SQL ETL pipelines to clean and load administrative health datasets into PostgreSQL, cutting nightly load time from about 6 hours to 2.
- Developed containerized microservices behind an API gateway to provide audited, access-controlled data extracts to approved researchers.

University of British Columbia, Department of Computer Science

Vancouver, BC

Teaching Assistant (CPSC 110, CPSC 121)

January 2023 – April 2024

- Led weekly tutorials of about 30 students on functional programming and discrete math, with a focus on recursion and proof techniques.
- Graded assignments and exams for a 300-student course and gave written feedback on proofs and program design.

Oil and Natural Gas Corporation

New Delhi, IN

Summer Intern

Summer 2022, Summer 2023

- Analyzed historical pipeline sensor data in Python to flag pressure anomalies that preceded past maintenance events, and presented candidate early-warning thresholds to the engineering team.
- Set up a database to track roughly 1,000 equipment inspection records and maintenance schedules.

RESEARCH EXPERIENCE

Carnegie Mellon University

Pittsburgh, PA

Undergraduate Research Assistant

May 2025 – July 2025

- Studied compositional verification techniques for concurrent programs as part of a faculty-led research project.
- Ran static-analysis tools on open-source multithreaded C/C++ code to find candidate data races.

UBC Pacific Laboratory of Artificial Intelligence

Vancouver, BC

Data Engineering Research Assistant

September 2023 – April 2024

- Collected and annotated Minecraft gameplay data and built Python preprocessing pipelines to structure it for behavior-modeling research.

EDUCATION

University of British Columbia

Vancouver, BC

Bachelor of Applied Science in Computer Engineering & Honours Mathematics

2021 – 2026